

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

Duration :1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

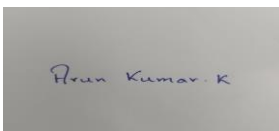
CASE STUDY

Code of Conduct:

I Kurangi Arun Kumar/1923722287 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Arun Kumar K

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Answer :

Case Study 1: Smart Medical Diagnosis System

1. Propositional Logic Representation

Propositional Symbols

- FS : Patient A has Fever
- CS : Patient A has Cough
- RS : Patient A has Rash
- BS : Patient A has Breathlessness
- $FluS$: Patient A has Possible Flu
- MS : Patient A has Possible Measles
- PS : Patient A has Possible Pneumonia

Knowledge Base (KB)

- **Rules:**
 - $R_1: (F \wedge C) \rightarrow Flu$
 - $R_2: (F \wedge R) \rightarrow MS$
 - $R_3: (C \wedge B) \rightarrow PS$
- **Facts:**
 - $F_1: FS$
 - $F_2: CS$
 - $F_3: BS$

2. Forward Chaining

Forward chaining begins with known facts and applies Modus Ponens to derive new facts iteratively.

- **Initial Known Facts:** $\{F, C, B\}$
- **Step 1:**
 - Check R_1 : Antecedent $(F \wedge C)$ is satisfied because F and C are in the known facts set.
 - **Rule Fired:** R_1
 - **Fact Derived:** Flu
 - **Updated Known Facts:** $\{F, C, B, Flu\}$
- **Step 2:**
 - Check R_2 : Antecedent $(F \wedge R)$ is not satisfied (R is missing).
 - Check R_3 : Antecedent $(C \wedge B)$ is satisfied because C and B are in the known facts set.
 - **Rule Fired:** R_3
 - **Fact Derived:** P
 - **Updated Known Facts:** $\{F, C, B, Flu, P\}$

Derived Diagnoses for Patient A: Possible Flu (Flu) and Possible Pneumonia (P).

3. Backward Chaining & Goal Reduction Tree

- **Goal:** P (Patient A has Pneumonia)



1. **Goal:** Prove P .
2. Match P with the consequent of R_3 : $(C \wedge B) \rightarrow P$.
3. Set subgoals: Prove C and prove B .
 - a. Subgoal C matches Fact F_2 ($C = \text{True}$).
 - b. Subgoal B matches Fact F_3 ($B = \text{True}$).
4. Both subgoals succeed, verifying the goal $P = \text{True}$.

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Answer :

Case Study 2: Autonomous Traffic Management Agent

1. Unification and Most General Unifier (MGU)

- **Rule 1:** $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$
- **Fact 1:** $\text{Vehicle}(\text{Ambulance})$
- **Fact 2:** $\text{EmergencyType}(\text{Ambulance})$

Unification Steps

1. Unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$:

$$\theta_1 = \{x / \text{Ambulance}\}$$

2. Substitute $x = \text{Ambulance}$ into $\text{EmergencyType}(x)$, yielding $\text{EmergencyType}(\text{Ambulance})$.

3. Unify $\text{EmergencyType}(\text{Ambulance})$ with Fact 2:

$$\theta_2 = \{\}$$

4. **Combined MGU (θ):**

$$\theta = \{x / \text{Ambulance}\}$$

Applying θ to Rule 1 yields:

$$\text{ClearPath}(\text{Ambulance})$$

2. Resolution Proof for Proceed(CarA)

Knowledge Base Clauses in Conjunctive Normal Form (CNF)

Converting rules to CNF by replacing $A \rightarrow B$ with $\neg A \vee B$:

1. $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
2. $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$
3. $\neg \text{ClearPath}(y) \vee \neg \text{RedSignal}(y)$
4. $\neg \text{Vehicle}(z) \vee \neg \text{Behind}(z, w) \vee \neg \text{GreenSignal}(w) \vee \text{Proceed}(z)$
5. $\text{Vehicle}(\text{Ambulance})$
6. $\text{EmergencyType}(\text{Ambulance})$

7. SC_7 : $\neg \text{Vehicle}(\text{CarA})$
8. SC_8 : $\text{Behind}(\text{CarA}, \text{Ambulance})$
9. **Negated Goal (SC_9):** $\neg \text{Proceed}(\text{CarA})$

Step-by-Step Resolution Steps

1. **Resolve SC_4 and SC_9** using substitution $\theta = \{z / \text{CarA}\}$:
 - a. SC_{10} : $\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, w) \vee \neg \text{GreenSignal}(w)$
2. **Resolve SC_{10} and SC_7 :**
 - a. SC_{11} : $\neg \text{Behind}(\text{CarA}, w) \vee \neg \text{GreenSignal}(w)$
3. **Resolve SC_{11} and SC_8** using substitution $\theta = \{w / \text{Ambulance}\}$:
 - a. SC_{12} : $\neg \text{GreenSignal}(\text{Ambulance})$
4. **Resolve SC_{12} and SC_2** using substitution $\theta = \{y / \text{Ambulance}\}$:
 - a. SC_{13} : $\neg \text{ClearPath}(\text{Ambulance})$
5. **Resolve SC_{13} and SC_1** using substitution $\theta = \{x / \text{Ambulance}\}$:
 - a. SC_{14} : $\neg \text{Vehicle}(\text{Ambulance}) \vee \neg \text{EmergencyType}(\text{Ambulance})$
6. **Resolve SC_{14} and SC_5 :**
 - a. SC_{15} : $\neg \text{EmergencyType}(\text{Ambulance})$
7. **Resolve SC_{15} and SC_6 :**
 - a. SC_{16} : $\square$ (Empty Clause)

Since the empty clause $\square$ is derived, the negated goal leads to a contradiction. Hence, $\text{Proceed}(\text{CarA})$ is True.

3. Analysis: Handling Simultaneous Emergency Vehicles

The current FOL KB is **insufficient** to handle two simultaneous emergency vehicles without conflict.

Key Limitations

- **Lack of Prioritization:** Rules treat all emergency vehicles identically ($\forall x$). If two emergency vehicles intersect or travel behind one another, there is no logic to resolve rights-of-way.
- **Static Signal Control:** Rule 2 assigns green signals globally to any path clearing an emergency vehicle without checking for intersecting directions, which could turn signals green simultaneously across perpendicular roads.

Required Additions

1. **Priority Hierarchy Function:** Introduce a priority mapping based on vehicle type or severity (e.g., $\text{Priority}(x) > \text{Priority}(y)$).
2. **Conflict Resolution Rule:**

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Answer :

Case Study 3: Agricultural Expert Reasoning System

1. CNF Conversion

Symbols

- \$SD\$: SoilDry
- \$IN\$: IrrigationNeeded
- \$CW\$: CropWheat
- \$ADM\$: ApplyDripMethod
- \$CAR\$: CropAtRisk

Step-by-Step CNF Transformations

- **P1:** $\text{SD} \rightarrow \text{IN}$
 - Eliminate Implication: $\neg \text{SD} \vee \text{IN}$
 - **CNF 1:** $\neg \text{SD} \vee \text{IN}$
- **P2:** $(\text{IN} \wedge \text{CW}) \rightarrow \text{ADM}$
 - Eliminate Implication: $\neg(\text{IN} \wedge \text{CW}) \vee \text{ADM}$
 - Move $\neg$ inward (De Morgan's Law): $\neg \text{IN} \vee \neg \text{CW} \vee \text{ADM}$
 - **CNF 2:** $\neg \text{IN} \vee \neg \text{CW} \vee \text{ADM}$
- **P3:** $(\neg \text{ADM} \wedge \text{CW}) \rightarrow \text{CAR}$
 - Eliminate Implication: $\neg(\neg \text{ADM} \wedge \text{CW}) \vee \text{CAR}$
 - Move $\neg$ inward: $\neg(\neg \text{ADM}) \vee \neg \text{CW} \vee \text{CAR}$
 - Double Negation Elimination: $\text{ADM} \vee \neg \text{CW} \vee \text{CAR}$
 - **CNF 3:** $\text{ADM} \vee \neg \text{CW} \vee \text{CAR}$
- **Facts:**
 - **CNF 4:** SD

- **CNF 5:** SCW

2. Resolution Refutation for ApplyDripMethod (SADM)

- **Goal:** SADM
- **Negated Goal:** $\neg \text{ADM}$ (CNF 6)

Set of Clauses

1. $\neg \text{SD} \vee \text{IN}$
2. $\neg \text{IN} \vee \neg \text{CW} \vee \text{ADM}$
3. $\text{ADM} \vee \neg \text{CW} \vee \text{CAR}$
4. SD
5. CW
6. $\neg \text{ADM}$

Resolution Proof Tree

$$\begin{array}{rcl}
 (2) \neg \text{IN} \vee \neg \text{CW} \vee \text{ADM} & (6) \neg \text{ADM} & \\
 \hline
 (7) \neg \text{IN} \vee \neg \text{CW} & (5) \text{CW} & \\
 \hline
 (8) \neg \text{IN} & (1) \neg \text{SD} \vee \text{IN} & \\
 \hline
 (9) \neg \text{SD} & (4) \text{SD} & \\
 \hline
 (10) \square & &
 \end{array}$$

Step-by-Step Derivation

1. Resolve (2) $\neg \text{IN} \vee \neg \text{CW} \vee \text{ADM}$ and (6) $\neg \text{ADM}$ $\rightarrow$ (7) $\neg \text{IN} \vee \neg \text{CW}$
2. Resolve (7) $\neg \text{IN} \vee \neg \text{CW}$ and (5) CW $\rightarrow$ (8) $\neg \text{IN}$
3. Resolve (8) $\neg \text{IN}$ and (1) $\neg \text{SD} \vee \text{IN}$ $\rightarrow$ (9) $\neg \text{SD}$
4. Resolve (9) $\neg \text{SD}$ and (4) SD $\rightarrow$ (10) $\square$

Since the empty clause $\square$ is derived, the theorem **ApplyDripMethod (SADM) is proven true.**

3. Step-by-Step Analysis when SoilDry is Removed

Knowledge Base Changes

- Removed Fact: SD
- Remaining KB: $\{ P1: \text{SD} \rightarrow \text{IN}, \quad P2: (\text{IN} \wedge \text{CW}) \rightarrow \text{ADM}, \quad P3: (\neg \text{ADM} \wedge \text{CW}) \rightarrow \text{CAR}, \quad \text{Fact}: \text{CW} \}$

Impact Evaluation

1. **Status of IrrigationNeeded (SIN):**

- a. $\$SD\$$ is no longer known to be true.
- b. $\$P1\$$ requires $\$SD = \text{True}\$$ to apply Modus Ponens. $\$IN\$$ becomes **unknown**.

2. Status of ApplyDripMethod ($\$ADM\$$):

- a. $\$P2\$$ requires both $\$IN\$$ and $\$CW\$$ to be true.
- b. Since $\$IN\$$ is unknown, $\$ADM\$$ cannot be derived as true ($\$ADM\$$ is **unknown**).

3. Status of CropAtRisk ($\$CAR\$$):

- a. $\$P3\$$ states $\$(\neg ADM \wedge CW) \rightarrow CAR\$$.
- b. To trigger $\$P3\$$, we need both $\$\neg ADM = \text{True}\$$ and $\$CW = \text{True}\$$.
- c. Although $\$CW = \text{True}\$$, $\$ADM\$$ is **unknown** (neither proved true nor proved false under standard monotonic propositional logic).
- d. Since $\$ADM\$$ is unknown, $\$\neg ADM\$$ cannot be inferred as true.
- e. Therefore, **CropAtRisk ($\$CAR\$$) DOES NOT follow**.

Reasoning Pattern Justification

Under **Monotonic Propositional Logic**, truth assignments follow strict sound inferences:

- Without $\$SD\$$, $\$IN\$$ is unassigned.
- The system uses the Open-World Assumption (OWA): an unproven statement is unknown, not automatically false.
- Without establishing $\$\neg ADM\$$, the antecedent for $\$P3\$$ remains unfulfilled, meaning $\$CAR\$$ cannot be logically entailed.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

Answer :x

Case Study 4 — Wumpus World Knowledge Agent

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$\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand